



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECV2113

HUMAN COMPUTER INTERACTION

ASSIGNMENT 1

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(a) One powerful example of a technology that advances the Sustainable Development Goals (SDGs), specifically Goal 4: Quality Education, is Duolingo. Its goal is to make easily accessible language learning materials available to anyone everywhere, assisting in the removal of linguistic obstacles and fostering international understanding and communication.

This is how Duolingo functions: It provides free language instruction in more than thirty languages, from widely spoken languages like French and Spanish to less spoken languages like Navajo and Swahili. Points, streaks, and levels are gamification strategies used by the site to encourage users to keep studying. To improve language learning, it also adjusts to users' learning preferences and offers tailored feedback. The app also uses friendly notifications to motivate user to learn the language proficiently and practice languages daily.

b)

1) Gamification Enhancing User's Motivation

Duolingo uses friendly notifications and reminders to encourage users to practice languages daily, sparking a sense of hope and enthusiasm for language learning. These reminders act as gentle nudges that prompt users to engage with the fun language learning activities offered by the app, igniting a spark of motivation to improve their language skills. By sending these reminders, Duolingo helps users build a habit of practicing languages regularly, turning language learning into an exciting part of their daily routine filled with hope for progress.

The fun and playful elements in Duolingo, such as earning points and interacting with cute characters, further enhance users' motivation and spark their interest in learning. These gamified features make language learning enjoyable, reducing the perceived difficulty and instilling a sense of hope that users can achieve their language goals. As a result, users are more motivated to spend time on the app and engage consistently with the language lessons, fueled by the hope of mastering new skills.

Duolingo's gamified approach not only makes language learning feel like a game but also fosters a sense of progress and achievement, reinforcing users' hope and motivation. The interactive nature of the app, combined with clear progression through levels and achievements, allows users to track their progress and celebrate their accomplishments. This sense of progress further motivates users to continue learning and fuels their commitment to mastering languages through Duolingo's engaging tools

2) Notifications Facilitating Engagement and Reminding users

Duolingo's notifications serve as gentle reminders that prompt users to return to the app for their daily language practice, acting as both a motivator and a facilitator in the language learning journey. These reminders are strategically designed to encourage users to stay consistent with their language learning routines, instilling a sense of fear of losing streaks on progress and improvement. By sending notifications at appropriate times, Duolingo aims to keep language learning on users' minds and encourage them to engage regularly with the app's language lessons and activities, acting as a facilitator for their learning process.

Initially, these notifications play a crucial role in motivating users to engage regularly with the app. They serve as positive reinforcement, reminding users of their language learning goals and encouraging them to take proactive steps towards achieving those goals, thus alleviating their fear of falling behind. However, over time, if notifications become too frequent, repetitive, or

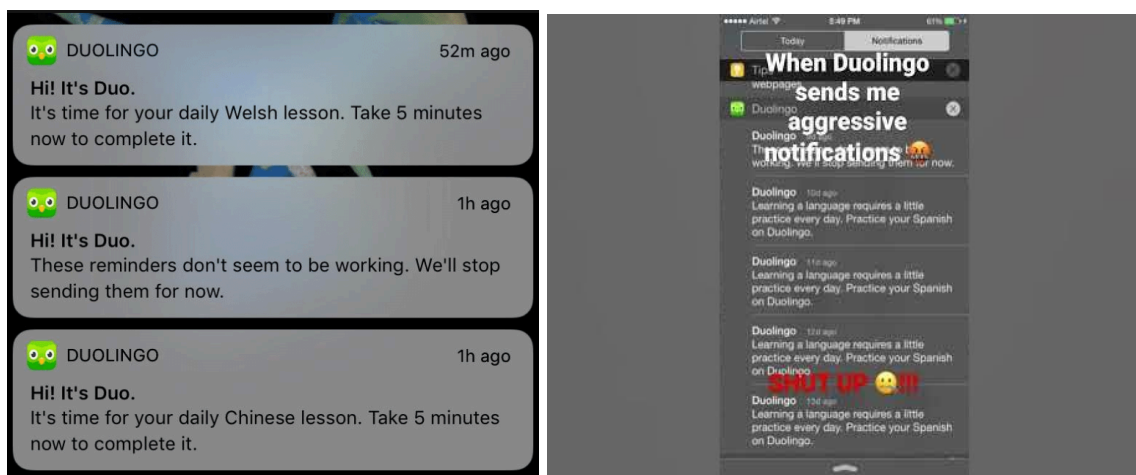
intrusive, they can have the opposite effect and lead to annoyance and stress, triggering a fear of interruptions rather than helpful reminders.

To effectively leverage notifications for user engagement, Duolingo must strike a balance between frequency and content, acting as a facilitator for learning while managing users' fear of notification overload. Well-timed and personalized notifications can effectively remind users to practice languages without causing stress or annoyance. By understanding users' preferences and learning habits, Duolingo can tailor notifications to provide relevant and meaningful reminders, thereby acting as a facilitator and easing users' fears of missing out or feeling overwhelmed. Balancing the frequency of notifications to avoid overwhelming users while ensuring they remain informed and encouraged is essential to maintaining users' ability to engage consistently with the app and achieve their language learning objectives.

d) The notification system is one of the Duolingo feature that might not be able to convince users to carry out the desired behaviour. Notifications should be used as gentle reminders to users to practise their language skills on a daily basis, but if they are excessively frequent, repetitive, or obtrusive, they may lose their effectiveness. The Fogg Behaviour Model (FBM) can be utilised for the analysis of this matter.

The FBM states that three things affect behaviour: ability, motivation, and triggers. In the context of Duolingo's notification, factors interaction are examined.

1. **Motivation:** The desire to learn, the gamified aspects of the software, and the sense of accomplishment it offers may initially be what push users to use Duolingo and practise languages on a daily basis. On the other hand, overbearing or pointless notifications might eventually sap motivation by irritating consumers or leading to notification fatigue. Notifications can decrease motivation over time by annoying users or causing notification fatigue.
2. **Ability:** Users may easily participate in language classes and activities with Duolingo because to its user-friendly and accessible design. But if alerts interfere with users' work or show up at odd times, they might think using the app is harder than it has to be, which would make them less likely to reply to alerts.
3. **Triggers:** Users are reminded to interact with Duolingo through notifications. While current and relevant notifications might encourage users to access the app and practise their language skills, badly designed or excessively frequent notifications run the risk of being ignored or rejected, which defeats the purpose of successfully inducing the desired behaviour.



Example of too much notifications

The following suggestions are meant to enhance the notification feature of Duolingo and make it more successful in convincing users to carry out the desired behaviour, which is daily language practice:

1. **Personalisation:** Adapt alerts to the interests and study habits of individual users. Permit users to personalise their notification settings according to their preferred language study time, desired reminder frequency, and most useful notification types.
2. **Timing Optimisation:** By analysing user behaviour patterns with machine learning algorithms, you may find the best times to send alerts based on what users are most likely to interact with the app. Remain clear of notifying users when they are likely to be busy or inactive.
3. **Variety and Relevance:** Make sure that alerts correspond with the objectives and advancement of users' language learning. Provide users personalised prompts based on their ongoing courses, approaching tasks, or goals obtained within the app, as an alternative to generic reminders. Make sure your notification content is varied to keep users interested and involved.
4. **Observe User Preferences:** Provide simple, clear choices in the app settings for controlling notification preferences. Give users complete control over their notification experience by enabling them to quickly change the timing, frequency, and types of notifications they get.

By putting these suggestions into practice, Duolingo can improve the effectiveness of its alert system and assist users with greater success in their language learning attempts without bothering or overwhelming them. This strategy respects user autonomy, relevance, and modification that is user-centered, all of which are in line with the principles of requiring design.

